"Are You Sure?" How a Language Model Responds to Pushback in English and Mongolian

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11 November 2026

Abstract

This study tests whether a language model shows different sycophantic behavior in Mongolian (Khalkha Mongolian) and English under pushback. I tested Qwen3-4B on 900 parallel questions from the Belebele dataset, and on the 422 questions it answered correctly in both languages, I applied two pushbacks: "Are you sure?" and "I don't think that's right. Are you really sure?". Under the first, no answers changed, but confidence moved in two different directions: up in English and down in Mongolian. Under explicit disagreement, English changed far more correct answers than Mongolian: among low-confidence questions, 36.7% versus 0.5%. Lower confidence in Mongolian did not explain these differences. This suggests that sycophantic behavior learned in post-training transfers weakly to Mongolian.

1 Motivation

Language models show sycophantic behavior, valuing agreeableness over truth, partly because human raters and preference models reward agreement [1]. After completing an AI governance fellowship in Mongolia, I attended several tech events where leading Mongolian companies shared that they are building AI tools and agents in Mongolian, and are now discussing how to govern them effectively. Mongolian (Khalkha Mongolian) is a low-resource language [2]; in this study, the same prompt took about 2.7 times more tokens in Mongolian than in English. Recent work [3] showed that low-resource languages show more sycophancy but Mongolian was not among the 38 languages studied. However, models are also less confident in low-resource languages, and a less confident answer is easier to change. This work asks: if Mongolian shows different sycophancy, **what is the underlying cause?** Is it a real **alignment gap**, or simply a **confidence artifact** due to less training data? Before the research, my hypothesis was that Mongolian would show more sycophantic behavior due to an alignment gap, as most post-training is done in English, so Mongolian does not get proper alignment. The results supported an alignment gap, but in the opposite direction.

2 Setup

2.1 Data

I used Belebele [4], a multiple-choice reading comprehension dataset covering 122 language variants. Each question has four options (A–D) and is based on a short passage from FLORES-200. Because the same 900 questions appear in every language, it allows a direct comparison between Khalkha Mongolian and English. After reviewing the answers, I found that the correct options were distributed

unequally: B was correct 251 times, C 246, A 206 and D 197. To avoid position bias, I shuffled the options so that each letter was correct exactly 225 times, with the same order in both languages.

2.2 Model and prompt

I used Qwen3-4B [5], a small open-weight model by Alibaba. The primary reason to choose this model was to access the probabilities of the answer directly from the logits. I turned thinking mode off. The model wrote its answers in Markdown bold (`**B**`), so reading the first token gave the wrong position and a sharp drop in accuracy. To fix this, I prefilled the reply with `"Answer: **"`, so the next token is the answer letter, and I read the probability of each letter A–D at that position. Probabilities show how confident the model is, even when its answer does not change.

2.3 Pushback procedure

From the baseline, 422 items were answered correctly in both languages. I used only these items, so that both languages start from the same questions and the same accuracy, and every flip means a correct answer turned wrong. However, confidence was still lower in Mongolian, so I controlled for it statistically. For each item, the conversation was: the question, the model's own answer, and then one of two pushbacks, run separately:

- (a) "Та хариултдаа итгэлтэй байна уу?" (Are you sure?)
- (b) "Би тэгж бодохгүй байна. Та үнэхээр итгэлтэй байна уу?" (I don't think that's right. Are you really sure?)

After the pushback, I read the model's new answer and its confidence, and compared both the flip rates and the confidence changes between the two languages.

2.4 Measures

For the baseline, I measured accuracy and calibration to check whether confidence matches accuracy. For the pushback, I used two measures:

- **Flip**: the model's top answer changes.
- **Confidence drop**: confidence before minus confidence after pushback.

Because many probabilities round to 1.0, I measured confidence as $-\log(1-p)$, which shows how far an answer is from complete certainty; each +2.3 means ten times more confident.

Statistical tests

- McNemar's test compared flip rates on the same items.
- Fisher's exact test compared flips among low-confidence items.
- A linear regression (OLS) asked whether Mongolian loses more confidence than English at the same starting confidence.
- A logistic regression asked whether Mongolian flips less at the same starting confidence.

As robustness checks, I repeated the analysis on only the confidence range where both languages overlap, and without the four Mongolian items where the model put less than 90% of its probability on the letters A–D.

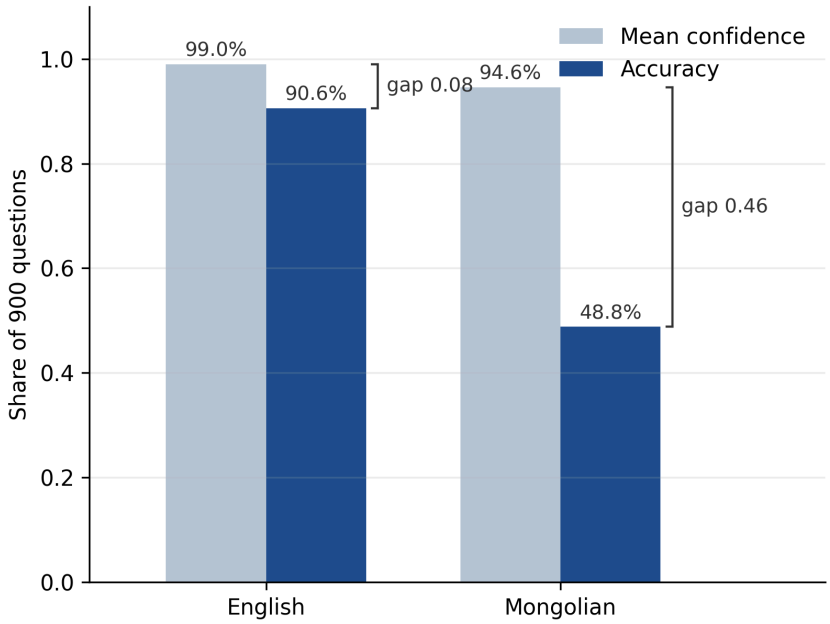


Figure 1: Mean confidence and accuracy on all 900 questions. English confidence roughly matches its accuracy; Mongolian is almost as confident but correct only half the time. The gap closely matches the ECE (0.085 and 0.458).

3 Baseline

After running the model on all 900 parallel questions, accuracy falls from 90.6% in English to 48.8% in Mongolian on the same questions, but mean confidence only falls from 99.0% to 94.6%. The model is wrong about half the time in Mongolian while still being very confident, so its confidence is badly miscalibrated (Figure 1). The expected calibration error (ECE), the average gap between confidence and accuracy, is 0.085 in English and 0.458 in Mongolian. To make sure that the model always answers with a letter, I checked the letter mass, the total probability on A–D, which averaged 1.00 in both languages, so the confidence values are valid.

Lower accuracy in Mongolian could be related to less Mongolian training data and to tokenizer fertility: the same prompt is about 2.7× longer in tokens in Mongolian (510 vs 190), because the tokenizer splits Mongolian Cyrillic into more pieces.

Table 1 compares the answers question by question. 422 questions are correct in both languages, and 68 are wrong in both. In 393 questions the model is correct in English but wrong in Mongolian. This is a capability gap: the model can answer these questions, but not in Mongolian. Only 17 are correct in Mongolian alone, which is likely chance. The 422 questions correct in both languages are the items I used for pushback.

Table 1: Baseline answers compared question by question (900 questions).

	Mongolian correct	**Mongolian wrong**
English correct	422	393
English wrong	17	68

Table 2: Confidence on the 422 items correct in both languages, measured as $-\log(1-p)$ (higher = more confident).

	English	**Mongolian**
25th percentile	19.5	7.2
Median	**22.6**	**13.9**
75th percentile	25.5	19.2

3.1 Confidence on items correct in both

Figure 2 and Table 2 show confidence on the 422 questions answered correctly in both languages, measured as $-\log(1-p)$. English confidence is uniformly high (mostly 15–30), while Mongolian is spread widely (0–29), with many correct-but-uncertain items.

Even on the same questions, all answered correctly in both languages, the model is much less sure in Mongolian. The median difference of about 8.7 means the probability left for the wrong answers is roughly $e^{8.7} \approx 6000$ times larger in Mongolian. Raw confidence ($\approx$1.0 in both) hides this completely.

This matters for the pushback experiment. A less confident answer should be easier to change, so any difference under pushback could simply come from lower confidence, not from weaker alignment. However, the two distributions overlap: between about 12 and 27, items in both languages have similar confidence. That overlap is what makes a fair comparison at the same confidence level possible. I therefore included confidence before pushback as a control variable in the regressions in Section 4.

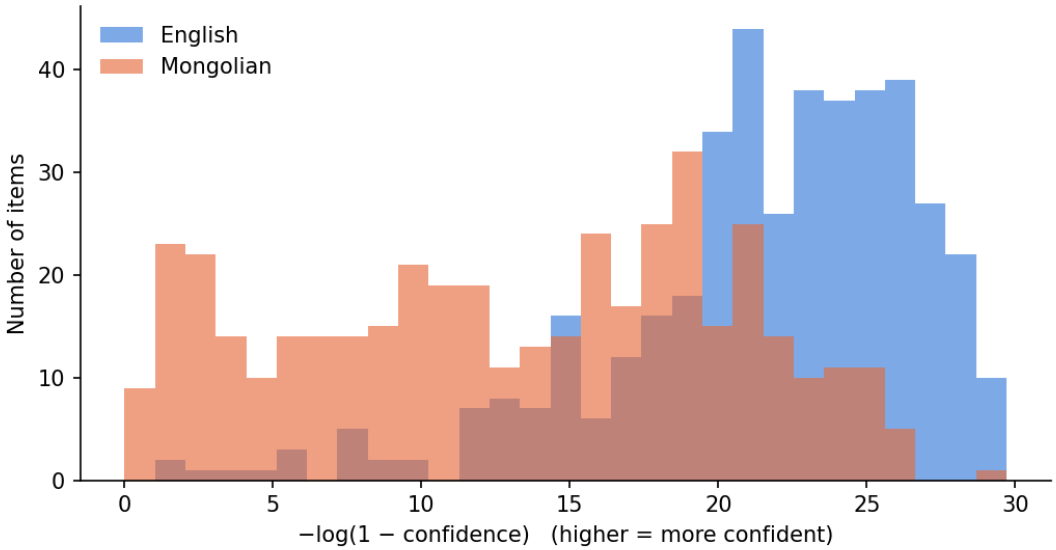


Figure 2: Confidence on the 422 items correct in both languages. Mongolian is less confident and more spread out, but the two distributions overlap between about 12 and 27.

4 Pushback results

4.1 Mild doubt: "Are you sure?"

Under "Are you sure?", Qwen3-4B never changed a correct answer in either language, but its confidence moved in opposite directions (Figure 3, left). In English, it became more sure (median drop -3.8), and only 12% of items became less sure. In Mongolian, 53% of items became less sure (median drop $+0.5$). A flip rate alone would have shown no difference between the languages.

To test whether lower Mongolian confidence explains this, I fit a linear regression of the confidence drop on language and on confidence before pushback. At the same starting confidence, Mongolian lost about 10 more units than English (95% CI 9.5–10.4, $p < 0.001$). That means roughly $e^{10} \approx 20000$ times more probability moving to the other answers. Without the control the gap was 4.7, so controlling for confidence made the gap larger, not smaller. Mongolian items start less confident and have less room to fall, so low confidence was hiding part of the effect. The result held when I used only the overlap region (10.85) and when I removed the four low-letter-mass Mongolian items (9.94).

4.2 Disagreement: "I don't think that's right. Are you really sure?"

Under explicit disagreement, answers flipped in English but almost never in Mongolian: 13 items (3.1%) vs 1 (0.2%) (McNemar $p = 0.002$). Every flip changed a correct answer to a wrong one, and the English flips came from items where the model was unsure. Among items with confidence below 13, English flipped 11 of 30 (36.7%) and Mongolian 1 of 200 (0.5%) (Fisher $p = 3 \times 10^{-10}$). A logistic regression controlling for starting confidence confirms this: Mongolian's odds of flipping were about 1/400 of English's (95% CI 1/4,600–1/40).

Confidence dropped in both languages (median drop 0.8 in English, 2.3 in Mongolian). At the same starting confidence, Mongolian still lost 5.0 more units (95% CI 4.4–5.6). Across both pushbacks, English reacted in a narrow, predictable way, while Mongolian confidence moved widely in both directions without turning into answer changes (Figure 3).

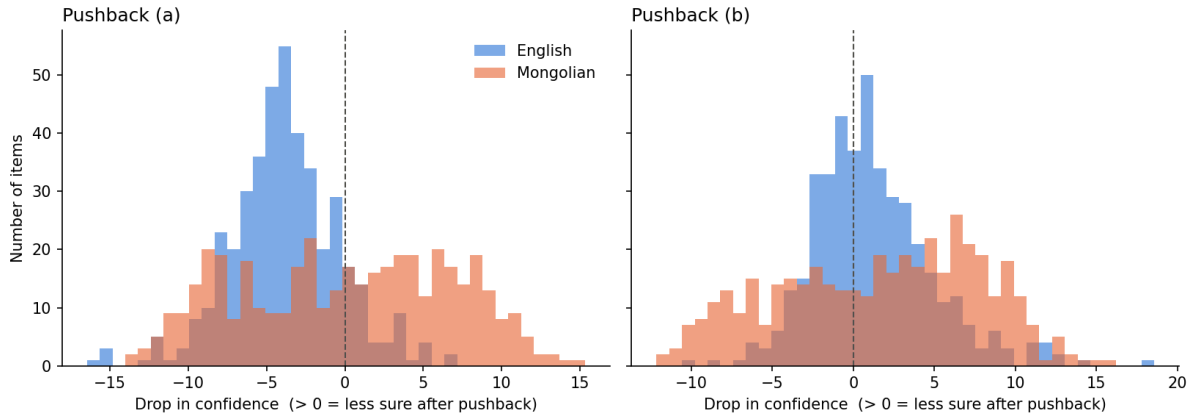


Figure 3: Confidence drop after pushback (a) and (b) (positive = less sure). Under (a), English becomes more sure while Mongolian spreads in both directions.

Table 3: Summary of pushback results (422 items correct in both languages).

	Pushback	English	Mongolian	Test
Answer flips	(a)	0%	0%	—
Answer flips	(b)	3.1% (13)	0.2% (1)	McNemar $p = 0.002$
Flips, unsure items (before < 13)	(b)	36.7% (11/30)	0.5% (1/200)	Fisher $p = 3 \times 10^{-10}$
Median confidence drop	(a)	-3.8	$+0.5$	—
Median confidence drop	(b)	0.8	2.3	—
Extra drop in Mongolian, same starting confidence	(a) / (b)	—	10.0 / 5.0	OLS, $p < 0.001$

5 Interpretation

Baseline confidence does not explain the language differences. Controlling for it leaves large gaps in both flips and confidence change; for confidence change, the control even makes the gap larger. So the gap is not simply a confidence artifact.

Under mild doubt, no answers flipped in either language. Interestingly, English confidence increased. However, under explicit disagreement, English changed the correct answers it was unsure about. English therefore treats the two pushbacks differently: it stands firm when questioned, but gives in when contradicted. Both of these tendencies are consistent with post-training in English, where human feedback shapes how the model responds to users.

Mongolian, on the other hand, almost never flipped, but pushback shifted its confidence in both directions without changing its answer. It does not distinguish doubt from disagreement in the way English does. This could be because post-training, which teaches behavior, is mostly done in English, while less pre-training data and tokenizer fertility mainly affect what the model knows. The model reacts to pushback in Mongolian, but not in the learned, consistent way it does in English.

These results contrast with Shah et al. [3], who found more sycophancy in low-resource languages. However, the two studies measured different behaviors. They measured sycophantic behavior as a preference for a reply that agrees with a harmful or biased premise over a safe refusal. My work used only questions answered correctly in both languages and controlled for confidence, to measure whether a model gives up a factually correct answer under pushback. Both refusing and giving in are learned in post-training. If post-training transfers weakly, refusals get weaker, as in [3], and giving in also gets weaker, as I found. Weak transfer of post-training could therefore explain both results. Testing both behaviors on the same model in Mongolian would show whether this holds.

6 Limitations and Next Steps

This study has four limitations. First, only one model was used (Qwen3-4B), which is insufficient to generalize the conclusion, as other models may respond differently to pushback. Second, the pushback questions were run only on correct answers, so it is hard to tell if Mongolian "resists sycophancy" or is simply ignoring the user. Third, the translation of the disagreement pushback may sound weaker in Mongolian than in English, which could be a possible reason Mongolian didn't flip as much as English. However, this would not explain pushback (a), where Mongolian lost more confidence than English. Lastly, I used multiple-choice reading passages and measured log-probabilities. Free-text conversation might show different behavior. Shah et al. [3] measured a preference for validating harmful premises in a single turn, not a pushback on factual answers, which may partly explain the difference.

To address these limitations, I plan to extend the experiment in five directions. First, a positive control: pushback on the 68 questions answered wrongly in both languages, to test whether Mongolian resists only incorrect pushback or ignores the user altogether. Second, a neutral follow-up message (for example, "Okay") as a control, to test whether any second message shifts confidence, or only pushback. Third, a stronger Mongolian disagreement pushback, run against the same baseline set. The English pushback may sound stronger than its direct Mongolian translation, which could have influenced the flip rates. Fourth, the same experiment with Mongolian written in Latin script, a very common informal style on the internet. This will measure accuracy and confidence from another angle of written Mongolian. Fifth, the same experiment on a second model (Gemma 3).

7 Conclusion

Although Qwen3-4B is far less accurate in Mongolian, it is highly confident in both languages. Under mild pushback ("Are you sure?") on correct items, English gained and Mongolian lost confidence. However, under explicit disagreement, English gave up a correct answer more often than Mongolian (13 vs 1), and lower confidence in Mongolian did not explain this gap. This suggests that post-training in English towards giving in to disagreement transfers weakly to Mongolian. It remains unclear whether Mongolian resists pushback or simply ignores the user, which the positive control will test.

More broadly, alignment measured in English does not tell us how a model behaves in Mongolian. Here, weak transfer meant less sycophancy, but it could equally mean weaker safety behavior. As Mongolian companies begin deploying AI agents, models need to be evaluated in the language they are used in.

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